

## # Push Down Automata (PDA):

PDA are enhanced version of F.A. A FA with a stack mpm can be viewed as PDA. Addition of stack mpm enhances the capability of PDA as compare to FA. Additional stack mpm is read+write mpm. The stack mpm is potentially infinite & its operation is based on last in first out. It means the last object pushed on the stack is popped first. It is just like array of plates kept together in the kitchen & we add or remove a plate from the top only. Stack mpm is also known as PD store. PDA accept the CPL.

## # Model of PDA:

A model of PDA is shown in fig. It consist of a finite tape, a reading head which reads from the tape, a stack mpm which is read/write and operated in last in first out fashion. There are two alphabets for input symbols to be read, one for input tape and another for stack. The stack alphabet is denoted by  $\Gamma$  and input alphabet is denoted by  $\Sigma$ . PDA reads from both alphabets, one symbol from the input tape and one symbol from top of the stack. After reading the inputs from tape & stack, PDA moves in next state & write 0 or more symbols on the stack.

## # Definition of PDA:

A PDA is described by 7 tuple  $(Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$  where,

- (1)  $Q$  is finite and non-empty set of states.
- (2)  $\Sigma$  is input alphabet.





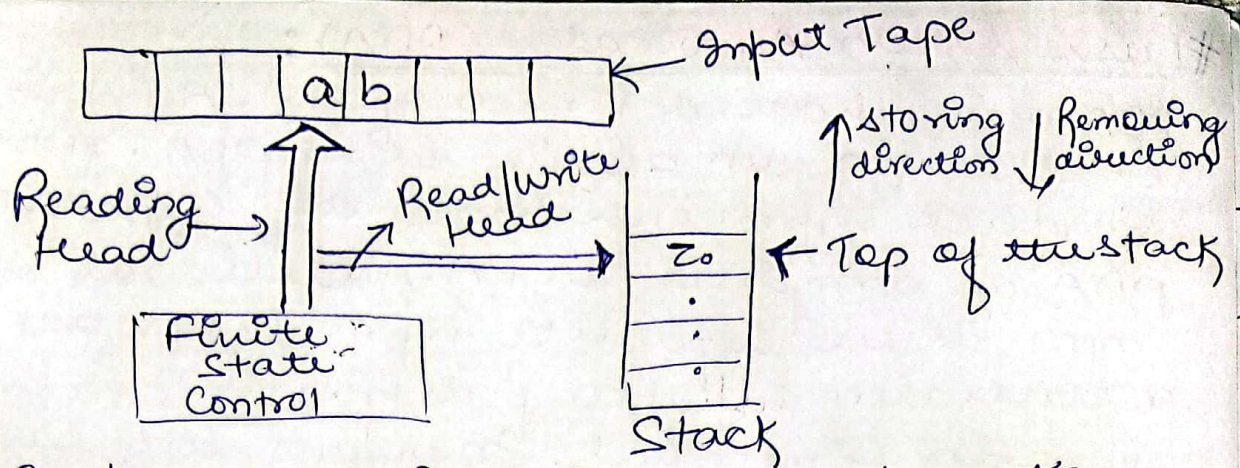


fig: Model of Push Down automata

- (3)  $\Gamma$  is finite and non-empty set of PD symbol.
- (4)  $\delta$  is the transition function, which maps

$$Q \times (\Sigma \cup \{\epsilon\}) \times \Gamma \rightarrow Q \times \Gamma^*$$

- (5)  $q_0 \rightarrow$  starting state,  $q_0 \in Q$
- (6)  $z_0 \in \Gamma$  is the starting (topmost or initial stack symbol)
- (7)  $F \subseteq Q$ , is the set of final states and sometime  $F = \emptyset$

### # Comparison b/w PDA and FA:

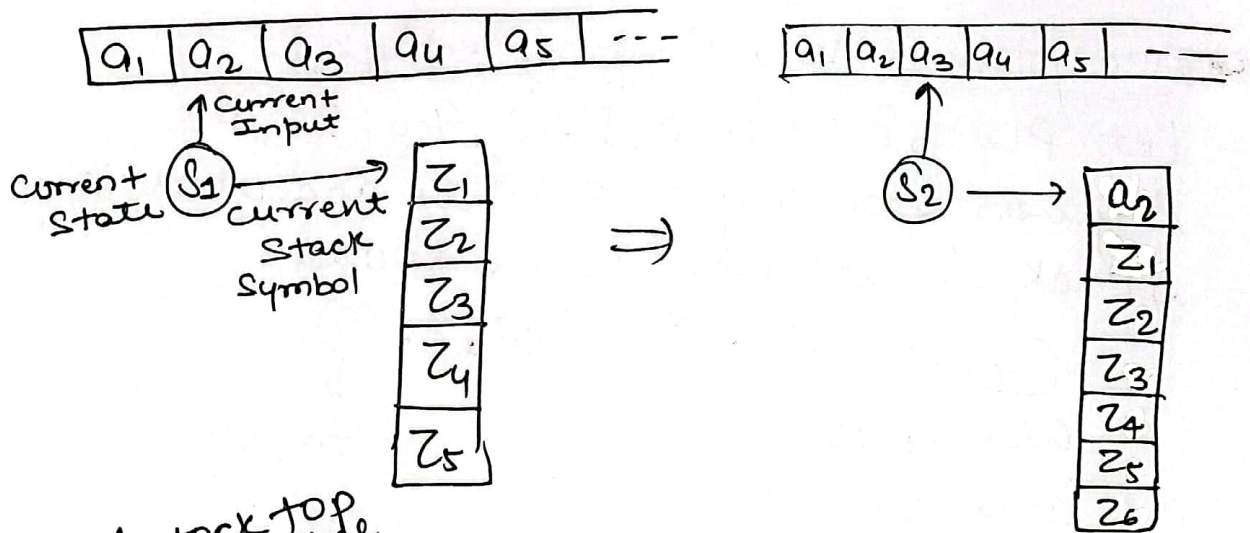
| PDA                                                                                              | FA                                                                                 |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| (1) The PDA can be defined as a collection of 7 tuple $(Q, \Sigma, \Gamma, \delta, z_0, F, q_0)$ | (1) The FA can be defined as a collection of 5 tuple $(Q, \Sigma, \delta, q_0, F)$ |
| (2) PDA can accept both Regular lang & CFL.                                                      | (2) FA can not accept CFL. FA can accept only Regular lang.                        |
| (3) PDA has a stack which is used to remember the I/P                                            | (3) FA has no stack storage.                                                       |
| (4) PDA is much more superior than FA                                                            | (4) FA is less powerful than PDA.                                                  |



# # Instantaneous Description (ID) of PDA:

## \* Definition:

let  $A = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$  be a PDA. An instantaneous description (ID) is  $(q, x, \alpha)$ , where  $q \in Q$ ,  $x \in \Sigma^*$  and  $\alpha \in \Gamma^*$ . For example  $(q, a_1 a_2 \dots a_n, z_1 z_2 \dots z_m)$  is an ID. This describes the PDA when the current state is  $q$ , the input string to be processed is  $a_1 a_2 \dots a_n$ . The PDA will process  $a_1, a_2 \dots a_n$  in that order. The PDA has  $z_1 z_2 \dots z_m$  with  $z_1$  at the top element of stack,  $z_2$  is the second element from the top, etc. and  $z_m$  is the lowest element in PDS (push down store).



current stack top symbol

$$\delta(q_0, x, z_0) = \delta(q_1, x z_0)$$

read input on the tape

change the state from  $q_0$  to  $q_1$

push  $x$  onto the stack

fig: Push operation.

$$\delta(q_0, x, z_0) = \delta(q_1, \epsilon)$$

current stack top symbol

change state from  $q_0$  to  $q_1$

pop the stack.

fig: Pop operation.

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## # Language of PDA:

There are two kinds of possible acceptance by PDA for any string -

(i) Acceptance by final state.

(ii) Acceptance by Empty stack.

### ● Acceptance by final state:

Let PDA,  $P = \{Q, \Sigma, \Gamma, \delta, q_0, z_0, F\}$

Let  $w \in \Sigma^*$  is a string, which is accepted by

PDA  $P$  by final state. It will be represented

as -  $(q_0, w, \gamma) \xrightarrow{*} (q_f, \epsilon, \gamma)$

where  $q_f \in F, \gamma \in \Gamma^*$

### ● Acceptance by Empty stack:

Let PDA,  $P = \{Q, \Sigma, \Gamma, \delta, q_0, z_0, F\}$

A given string  $w \in \Sigma^*$  is accepted by empty stack is defined as follows:

$(q_0, w, \gamma) \xrightarrow{*} (q_i, \epsilon, \epsilon)$

where,  $q_i \in F$

and  $\gamma \in \Gamma^*$

↓  
Empty  
Stack

Ques Construct PDA for language  $L = \{a^n b^n \mid n \geq 1\}$

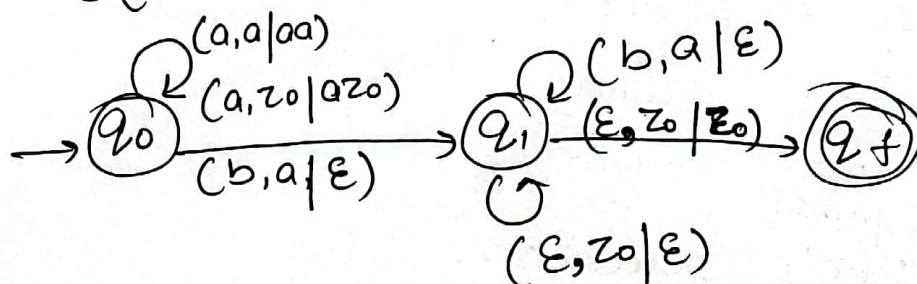
Sol<sup>n</sup>  $\delta(q_0, a, z_0) = (q_0, a z_0)$

$\delta(q_0, a, a) = (q_0, a a)$

$\delta(q_0, b, a) = (q_1, \epsilon)$

$\delta(q_1, b, a) = (q_1, \epsilon)$

$\delta(q_1, \epsilon, z_0) = (q_f, z_0) \text{ or } (q_1, \epsilon)$





Ques Construct PDA for lang  $L = \{ w \mid n_a(w) = n_b(w) \}$

Sol<sup>n</sup>  $L = \{ w \mid n_a(w) = n_b(w) \}$

$$\delta(q_0, a, z_0) = (q_0, az_0)$$

$$\delta(q_0, b, z_0) = (q_0, bz_0)$$

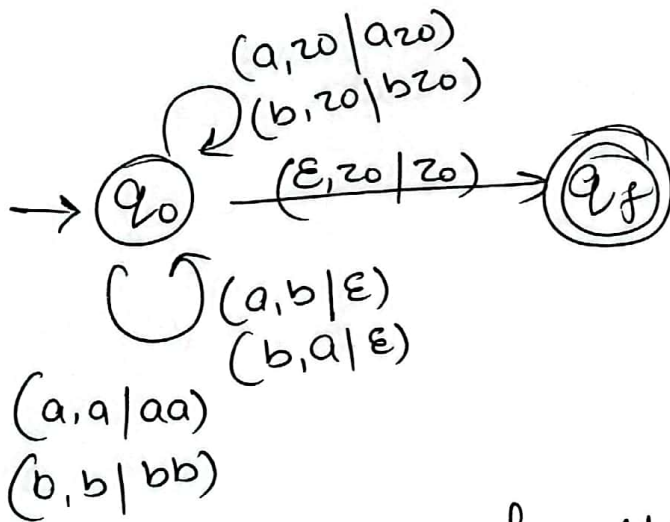
$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, b, b) = (q_0, bb)$$

$$\delta(q_0, a, b) = (q_0, \epsilon)$$

$$\delta(q_0, b, a) = (q_0, \epsilon)$$

$$\delta(q_0, \epsilon, z_0) = (q_f, z_0)$$



Ques Construct PDA for the lang.  $L = \{ a^n b^n c^m \mid n, m \geq 1 \}$

Sol<sup>n</sup>  $L = \{ a^n b^n c^m \mid n, m \geq 1 \}$

$$\delta(q_0, a, z_0) = (q_0, az_0)$$

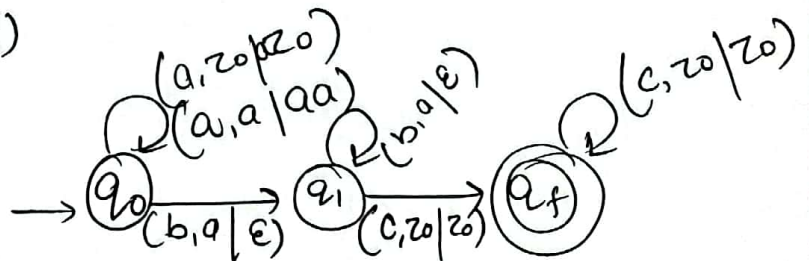
$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, c, z_0) = (q_f, z_0)$$

$$\delta(q_f, c, z_0) = (q_f, z_0)$$



(81)

Ques construct PDA for the language  $L = \{a^n b^m c^n \mid n, m \geq 1\}$

Sol<sup>n</sup>  $L = \{a^n b^m c^n \mid n, m \geq 1\}$

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

$$\delta(q_0, a, a) = (q_0, aa)$$

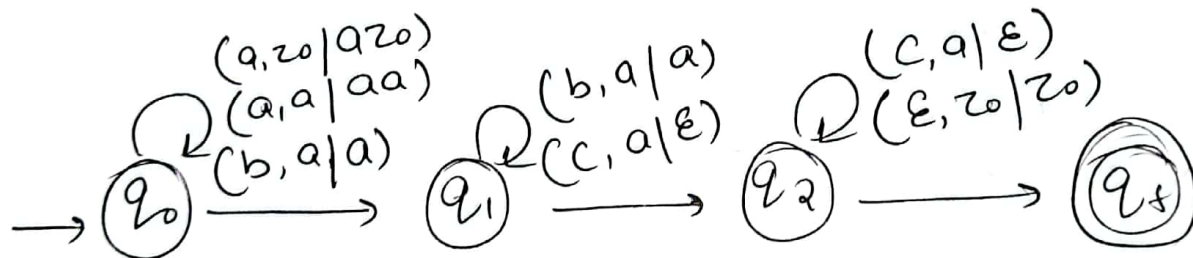
$$\delta(q_0, b, a) = (q_1, a)$$

$$\delta(q_1, b, a) = (q_1, a)$$

$$\delta(q_1, c, a) = (q_2, \epsilon)$$

$$\delta(q_2, c, a) = (q_2, \epsilon)$$

$$\delta(q_2, \epsilon, z_0) = (q_f, z_0)$$



Ques Construct PDA for the lang.  $L = \{a^n b^n \mid n \geq 1\}$

Sol<sup>n</sup>  $L = \{a^n b^n \mid n \geq 1\}$

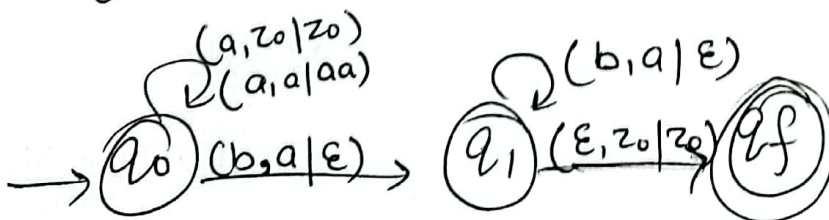
$$\delta(q_0, a, z_0) = (q_0, aa)$$

$$\delta(q_0, a, a) = (q_0, aaaa)$$

$$\delta(q_0, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_f, z_0)$$



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Ques construct PDA for the lang  $L = \{wcw^R \mid w \in (a,b)^+\}$

Sol<sup>n</sup>  $\delta(q_0, a, z_0) = (q_0, az_0)$

$\delta(q_0, b, z_0) = (q_0, bz_0)$

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, b) = (q_0, bb)$

$\delta(q_0, a, b) = (q_0, ab)$

$\delta(q_0, b, a) = (q_0, ba)$

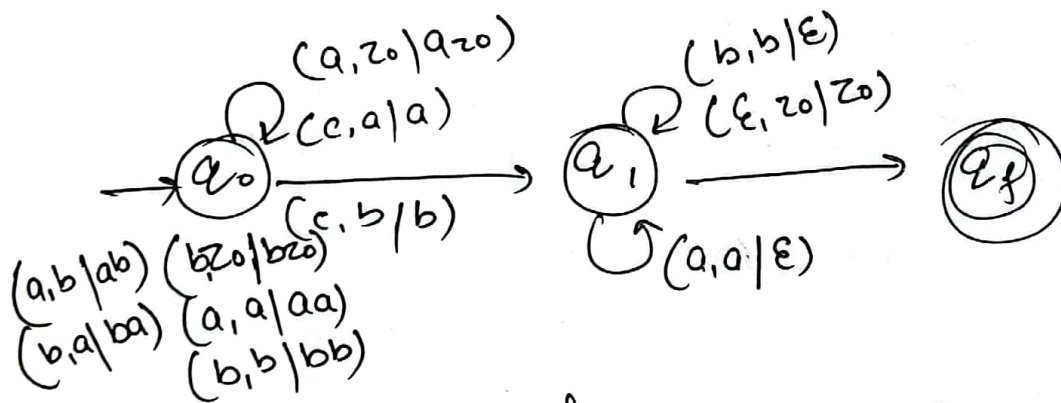
$\delta(q_0, c, a) = (q_1, a)$

$\delta(q_0, c, b) = (q_1, b)$

$\delta(q_1, a, a) = (q_1, \epsilon)$

$\delta(q_1, b, b) = (q_1, \epsilon)$

$\delta(q_1, \epsilon, z_0) = (q_f, z_0)$



Ques Construct PDA for lang.  $L = \{ww^R \mid w \in (a,b)^+\}$

Sol<sup>n</sup>  $\delta(q_0, a, z_0) = (q_0, az_0)$

$\delta(q_0, b, z_0) = (q_0, bz_0)$

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, b) = (q_0, bb)$

$\delta(q_0, a, b) = (q_0, ab)$

$\delta(q_0, b, a) = (q_0, ba)$

$\delta(q_0, a, a) = (q_1, \epsilon)$

$\delta(q_0, b, b) = (q_1, \epsilon)$

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$$\delta(q_2, a, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, b) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_f, z_0)$$

## # Content Free Grammar to Push Down Automata:

Let  $L = L(G)$ , where  $G = (V_N, \Sigma, P, s)$  is CFG. We construct

a PDA,  $A$  as  $A = (\{q\}, \Sigma, V_N \cup \Sigma, \delta, q, s, \phi)$

where  $\delta$  is defined by the following Rules:

$$R_1: \delta(q, \wedge, A) = \{(q, \alpha) \mid A \rightarrow \alpha \text{ is in } P\}$$

$$R_2: \delta(q, a, a) = \{(q, \wedge) \mid \text{for every } a \text{ in } \Sigma\}$$

\*  $\phi \rightarrow$  Empty stack

\*  $f \rightarrow$  for final state.

Ques Construct a PDA,  $A$  which is equivalent to the following given CFG.

$$S \rightarrow 0cc, S \rightarrow 0S, C \rightarrow 1S, C \rightarrow 0$$

Test whether  $010^4$  is accepted by  $N(A)$

Sol<sup>n</sup> The target PDA,  $A$  is as follows -

$$A \rightarrow (\{q\}, \{0, 1\}, \{S, C, 0, 1\}, \delta, q, s, \phi)$$

$\delta$  is defined by the following rules -

$$R_1: \delta(q, \wedge, S) = \{(q, 0cc)\}$$

$$\delta(q, \wedge, C) = \{(q, 0S), (q, 1S), (q, 0)\}$$

$$R_2: \delta(q, 0, 0) = \{(q, \wedge)\}$$

$$\delta(q, 1, 1) = \{(q, \wedge)\}$$

$$\begin{aligned} (q, 010^4, s) &\vdash (q, 010^4, 0cc) \vdash (q, 10^4, cc) \vdash (q, 10^4, sc) \\ &\vdash (q, 0^4, sc) \vdash (q, 0^4, 0ccc) \vdash (q, 0^3, ccc) \vdash (q, 0^3, 0cc) \\ &\vdash (q, 0^2, cc) \vdash (q, 0^2, 0c) \vdash (q, 0, c) \vdash (q, 0, 0) \vdash \\ &(q, \wedge) \end{aligned}$$

Thus,  $010^4 \in N(A)$   $\text{Ans}$

TH



$$S \Rightarrow 0cc$$

$$\Rightarrow 01sc \text{ (using } c \rightarrow 1s)$$

$$\Rightarrow 010ccc \text{ (using } s \rightarrow 0ccc)$$

$$\Rightarrow 010000 \text{ (using } c \rightarrow 0)$$

$$\Rightarrow 010^4$$

Ques Construct a PDA for the CFG given below.

$$S \rightarrow asb$$

$$S \rightarrow a|b|\epsilon$$

Sol<sup>n</sup>

$$S \rightarrow asb$$

$$S \rightarrow a|b|\epsilon$$

$$R_1: (q, \wedge, A) = \{(q, \alpha) \mid A \rightarrow \alpha \text{ is in } P\}$$

$$R_2: (q, a, a) = \{(q, \wedge)\} \text{ for every } a \in \Sigma$$

$$R_1: (q, \wedge, s) = \{(q, asb)\}$$

$$(q, s) = \{(q, a), (q, b), (q, \epsilon)\}$$

$$R_2: (q, a, a) = \{(q, \wedge)\}$$

$$(q, b, b) = \{(q, \wedge)\}.$$

Ques Construct a PDA for the CFG given below.

$$S \rightarrow 0s1|A$$

$$A \rightarrow 1AO|s|\epsilon$$

Sol<sup>n</sup>

$$R_1: (q, \wedge, A) = \{(q, \alpha) \mid A \rightarrow \alpha \text{ is in } P\}$$

$$R_2: (q, a, a) = \{(q, \wedge)\} \text{ for every } a \in \Sigma$$

$$R_1: (q, \wedge, s) = \{(q, 0s1), (q, A)\}$$

$$(q, \wedge, A) = \{(q, 1AO), (q, s), (q, \epsilon)\}$$

$$R_2: (q, 0, 0) = \{(q, \wedge)\}$$

$$(q, 1, 1) = \{(q, \wedge)\}.$$

Ques Consider the context free Grammar -

$(\{S, A, B\}, \{a, b\}, P, S)$ , where production  $P$

are as follows -

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$$S \rightarrow aABB \mid aAA$$

$$A \rightarrow aBB \mid a$$

$$B \rightarrow bBB \mid b$$

Sol<sup>n</sup>  $R_1: (q, \Lambda, s) = \{(q, aABB), (q, aAA)\}$

$$(q, \Lambda, A) = \{(q, aBB), (q, a)\}$$

$$(q, \Lambda, B) = \{(q, bBB), (q, b)\}$$

$$R_2: (q, a, a) = \{(q, \Lambda)\}$$

$$(q, b, b) = \{(q, \Lambda)\}$$

## # Push Down Automata to Context Free Grammar

If  $A = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$  is a PDA then CFG is defined as  $G = (VN, \Sigma, P, S)$

(1) Construction of set of Non-terminal

$$VN = \{S\} \cup \{[q, z, q'] \mid q, q' \in Q, z \in \Gamma\}$$

(2) <i> S-Production:

$$S \rightarrow [q_0, z_0, q], q \in Q$$

<ii> for Pop operation:

$$\delta(q, a, z) \rightarrow (q', \Lambda)$$

$$[q, z, q'] \rightarrow a$$

<iii> for Push and No operation:

$$\delta(q, a, z) \rightarrow (q_1, z_1 z_2 \dots z_m)$$

$$[q, z, q'] \rightarrow a[q_1, z_1, q_2][q_2, z_2, q_3] \dots [q_m, z_m, q']$$

where  $q', q_1, q_2, q_3, \dots, q_m \in Q$

Ques Construct CFG from PDA,  $A = (\{q_0, q_1\}, \{a, b\}, \Sigma, [z_0, z], \delta, q_0, z_0, \phi)$

$\delta:$

$$\delta(q_0, b, z_0) = (q_0, z z_0)$$

$$\delta(q_0, b, z) = (q_0, z z)$$

$$\delta(q_1, b, z) = (q_1, \Lambda)$$

$$\delta(q_0, \Lambda, z_0) = (q_0, \Lambda)$$

$$\delta(q_0, a, z) = (q_1, z)$$

$$\delta(q_1, a, z_0) = (q_0, z_0)$$

(ab)



$$\delta(1) N = \{ S, [q_0, z_0, q_0], [q_0, z_0, q_1], [q_0, z, q_0], [q_0, z, q_1], [q_1, z_0, q_0], [q_1, z_0, q_1], [q_1, z, q_0], [q_1, z, q_1] \}$$

$$\textcircled{2} \quad \delta \rightarrow [q_0, z_0, q_0]$$

$$S \rightarrow [q_0, z_0, q_1]$$

$$\textcircled{3} \quad \delta(q_0, b, z_0) = (q_0, zz_0) \text{ (push op.)}$$

$$[q_0, z_0, q_0] \rightarrow b[q_0, z, q_0][q_0, z_0, q_0]$$

$$[q_0, z_0, q_0] \rightarrow b[q_0, z, q_1][q_1, z_0, q_0]$$

$$[q_0, z_0, q_1] \rightarrow b[q_0, z, q_0][q_0, z_0, q_1]$$

$$[q_0, z_0, q_1] \rightarrow b[q_0, z, q_1][q_1, z_0, q_1]$$

$$\textcircled{4} \quad \delta(q_1, b, z) = (q_1, \wedge)$$

$$[q_1, z, q_1] \rightarrow b$$

$$\textcircled{5} \quad \delta(q_0, \wedge, z_0) = (q_0, \wedge) \text{ (pop op.)}$$

$$[q_0, z_0, q_0] \rightarrow \wedge$$

$$\textcircled{6} \quad \delta(q_0, a, z) = (q_1, z) \text{ (no op.)}$$

$$[q_0, z, q_0] \rightarrow a[q_1, z, q_0]$$

$$[q_0, z, q_1] \rightarrow a[q_1, z, q_1]$$

$$\textcircled{7} \quad \delta(q_1, a, z_0) = (q_0, z_0) \text{ (no op.)}$$

$$[q_1, z_0, q_0] \rightarrow a[q_0, z_0, q_0]$$

$$[q_1, z_0, q_1] \rightarrow a[q_0, z_0, q_1]$$

$$\textcircled{8} \quad \delta(q_0, b, z) = (q_0, zz)$$

$$[q_0, z, q_0] \rightarrow b[q_0, z, q_0][q_0, z, q_0]$$

$$[q_0, z, q_0] \rightarrow b[q_0, z, q_1][q_1, z, q_0]$$

$$[q_0, z_0, q_1] \rightarrow b[q_0, z, q_0][q_0, z, q_1]$$

$$[q_0, z, q_1] \rightarrow b[q_0, z, q_1][q_1, z, q_1]$$

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Ques let  $M = (\{q_0, q_1\}, \{0, 1\}, \{x, z_0\}, \delta, q_0, z_0, \phi)$   
where  $\delta$  is given by -

$$\delta(q_0, 0, z_0) = (q_0, xz_0)$$

$$\delta(q_0, 0, x) = (q_0, xx)$$

$$\delta(q_0, 1, x) = (q_1, \epsilon)$$

$$\delta(q_1, 1, x) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, x) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_1, \epsilon)$$

ES



## # Pumping lemma for CFL:

The pumping lemma for CFL states that for every sufficiently long string in CFL, there are always two short sub-strings close together that can be pumped same no. of times as the result is a string in the same CFL.

lemma:

Let  $L$  be a CFL and a long string  $w$  is in  $L$  then there exists a constant  $n$  such that  $|w| \geq n$  and  $w$  can be written as  $w = uvxyz$  such that:

- $\langle i \rangle |vxy| \geq 1$
- $\langle ii \rangle |vxy| \leq n$
- $\langle iii \rangle$  for all  $i \geq 0$   $uv^i xy^i z$  is in  $L$ .

Note:

[In this case  $x$  is not  $\epsilon$  and  $v$  and  $y$  are not both  $\epsilon$  at the same time].

Ques By using pumping lemma show that the language  $L = \{a^p \mid p \text{ is prime}\}$  is non-CFL.

Sol<sup>n</sup>  $L = \{a^p \mid p \text{ is prime}\}$

$p = 2, 3, 5, 7, 11, \dots$

$L = \{aa, aaa, aaaaa, \dots\}$

let  $w = aa$ .

we know that by pumping lemma,  
 $w = uv^i xy^i z$  for all  $i \geq 0$

let  $y = \epsilon$        $v = a$

$u = \epsilon$        $x = a$

$z = \epsilon$

QED

$$w = uv^i x y^i z$$

$$w = \varepsilon(a)^i a \varepsilon \varepsilon$$

$$w = (a)^i a$$

for  $i=1$ ,  $w = (a)^1 a = aa$ , which is in  $L$ .

for  $i=2$ ,  $w = (a)^2 a = aaa$ , which is in  $L$ .

for  $i=3$ ,  $w = (a)^3 a = aaaa$ , which is not in  $L$ .

Therefore,  $L = \{a^i \mid i \text{ is prime}\}$  is non-CFL.

Ques Prove that  $L = \{a^n b^n c^n \mid n > 0\}$  is not a CFL.

Sol<sup>n</sup>  $L = \{a^n b^n c^n \mid n > 0\}$

$$L = \{abc, aabbcc, aaabbbccc, \dots\}$$

$$\text{let } w = aabbcc$$

We know that by pumping lemma,

$$w = uv^i x y^i z \text{ for all } i \geq 0$$

$$u = \varepsilon \quad x = bb$$

$$y = \varepsilon \quad z = cc$$

$$v = aa$$

$$w = \varepsilon(aa)^i bb \varepsilon cc$$

$$w = (aa)^i bbcc$$

for  $i=1$ ,  $w = (aa)^1 bbcc = aabbcc$  which is in  $L$ .

for  $i=2$ ,  $w = (aa)^2 bbcc = aaabbbcc$ , which is <sup>not</sup> in  $L$ .

Therefore,  $L = \{a^n b^n c^n \mid n > 0\}$  is not a CFL.

Ques Show that  $L = \{a^i b^j \mid j = i^2\}$  is not a CFL.

Sol<sup>n</sup>  $L = \{a^i b^j \mid j = i^2\}$

$$L = \{ab, aabbbb, \dots\}$$

$$\text{let } w = ab$$

We know that by using pumping lemma,

$$w = uv^i x y^i z \text{ for all } i \geq 0$$

(20)



$$\begin{aligned} y &= \varepsilon \\ u &= \varepsilon \\ z &= \varepsilon \end{aligned}$$

$$\begin{aligned} v &= a \\ x &= b \end{aligned}$$

| $j$ | $i^2$ |
|-----|-------|
| 1   | 1     |
| 2   | 4     |
| 3   | 9     |

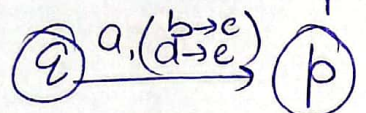
$$\begin{aligned} w &= v^i x \\ w &= a^i b \end{aligned}$$

for  $i=1$ ,  $w = a^1 b = a^1 b = ab$  which is in  $L$ .  
 for  $i=2$ ,  $w = a^2 b = a^2 b = aab$  which is not in  $L$ .

Therefore  $L = \{a^i b^j \mid j = i^2\}$  is not a CFL.

### # Two stack PDA:

A two stack PDA is similar to PDA but it has two stack. Instead of one. In each transition we must denote the pop and push on both stack



The diagram below means for input a:

- (1) b is popped off from the first stack and c is pushed on the first stack.
- (2) d is popped off from the second stack and e is pushed on the second stack.

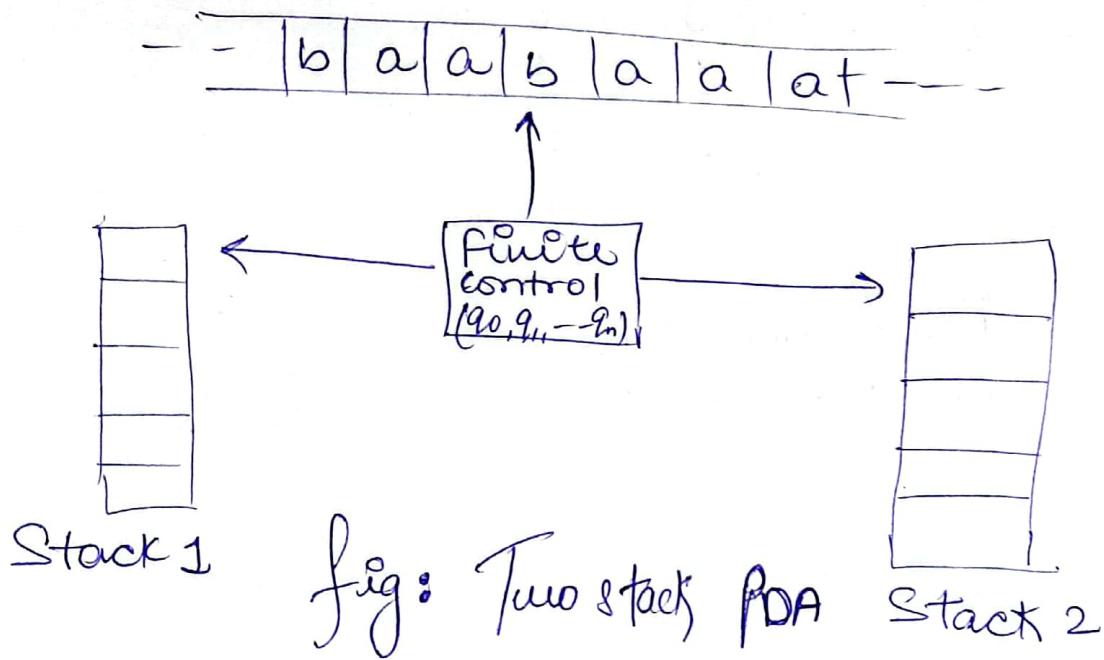
### # Definition:

A Two stack PDA is a 6-tuple  $(Q, \Sigma, \Gamma, \delta, q_0, F)$ , which  $Q, \Sigma, \Gamma, q_0$  and  $F$  are the same as in a one-stack PDA but the transition function  $\delta$  is different, which is as follows:

$$\delta: Q \times (\Sigma \cup \{\varepsilon\}) \times \Gamma \times \Gamma \rightarrow Q \times \Gamma^* \times \Gamma^*$$

Two stack PDA accept any language that accepted by a Turing m/c.

(a)



Q2